

Manoeuvre Pattern-of-Life Prediction for Space Domain Awareness

A sequential briefing on every model, method and result of the project

UROP project (Imperial College London) — briefing compiled from the code and its outputs

October 5, 2026

Abstract

This document explains, in the order in which the work was done, how a Starlink-like satellite that keeps its orbit inside a 500 m semi-major-axis (SMA) deadband was simulated, how its manoeuvres were detected from noisy tracking data, how the control law behind them was learned, and how that law was used to forecast future manoeuvres. It summarises the physics (two-body gravity, the J_2 oblateness term, atmospheric drag, lunar and solar third-body attraction, solar radiation pressure), the controller (a smooth *tanh* deadband with memory), the estimation methods (finite-difference regression, the energy method, shooting, Kalman filtering, spectral analysis), the classification methods (Gaussian mixture models and alternatives, scored per burn *event*), the learning of the control law (SINDy and a Bayesian variant) and nine extension studies aimed at the UK Space Agency brief (space weather, realistic tracking passes, conjunction screening, sensor tasking, multiple control laws, fleets, calibrated intervals, GEO and a library “zoo” for SINDy). Every number quoted comes from the latest run of `python main.py (outputs/results.json)`.

Contents

1	The brief and the question	4
1.1	How the work progressed	4
2	Units, reference orbit and numerical integration	5
2.1	Non-dimensional units	5
2.2	Reference orbit	5
2.3	Integrators	5
3	The dynamical model (report step 1)	5
3.1	Equations of motion	5
3.2	The force budget	7
3.3	The 400-day data sets	7
4	The station-keeping controller (report step 5)	7
4.1	Why the Week 4 controller had to change	7
4.2	The mean SMA	8
4.3	The smooth deadband law with memory	8
4.4	Results over 400 days	9

5	Step 0: bias–variance tuning of every setting	9
5.1	The decomposition	9
5.2	The studies and what they chose	10
5.3	Choosing the derivative by walk-forward and walk-backward testing	10
6	Step 2: learning the coefficients by regression	11
6.1	Finite differences	11
6.2	Linear least squares	11
7	Step 3: drag estimation and noise robustness	12
7.1	Three ways to get c_d	12
7.2	Step 3b: realistic (coloured) noise	13
7.3	Workstream E: whole-trajectory cost surface	13
8	Step 4: spectra, filtering and Kalman filtering	13
8.1	The periodogram and the J_2 signature	13
8.2	Filtering	14
8.3	The extended Kalman filter (EKF)	14
8.4	Step 4b: “force spectroscopy”	14
9	Step 6: classifying coast and burn	15
9.1	The feature: the unmodelled acceleration	15
9.2	The Gaussian mixture model	15
9.3	From samples to events	15
9.4	Results	15
9.5	Stress test: how coarse can the sampling get?	15
10	Step 7: robustness of the classifiers	16
11	Step 8: repetition — autocorrelation, stacking and a Gaussian process	17
12	Step 9: learning the control law (SINDy and BINDy) and forecasting	17
12.1	Sparse identification of nonlinear dynamics (SINDy)	17
12.2	The switching law	18
12.3	Forecasting	18
13	The workstream stages, aligned with the report data	19
14	Extensions towards the brief (steps 10–18)	19
14.1	The mean-element model	19
14.2	Step 10: space-weather drag	19
14.3	Step 11: realistic tracking passes	20
14.4	Step 12: close-approach screening	20
14.5	Step 13: sensor tasking and track association	21

14.6 Step 14: several control laws on one satellite	21
14.7 Step 15: a fleet of 40 satellites	21
14.8 Step 16: calibrated (conformal) burn-time intervals	21
14.9 Step 17: GEO station keeping	21
14.10Step 18: the SINDy equation zoo	21
15 Step 19: thrust law and natural dynamics, learned together or apart	22
15.1 Why split the problem?	22
15.2 The four scenarios	23
15.3 Results	23
15.4 The natural law with no deadband	24
16 Validation, figure audit and honest limitations	25
16.1 Testing	25
16.2 Problems found and fixed during the audits	25
16.3 Limitations	25
17 How well the project answers the brief	26
A Running the code	26
B Glossary	26

1 The brief and the question

The UK Space Agency brief “Manoeuvre Pattern of Life Prediction for SDA” starts from a simple observation: of the $\sim 50\,000$ tracked objects about 15 000 are active and may manoeuvre, and by 2035 mega-constellations will push this beyond 50 000. Unpredicted manoeuvres break three core Space Domain Awareness (SDA) functions:

1. **close-approach (conjunction) screening**, which relies on predicted trajectories being accurate over the screening window — in the brief’s Figure 1 a Starlink satellite keeping its SMA inside a 500 m deadband changes its miss distance by *hundreds of km in a week*;
2. **track association**, because a manoeuvre makes the predicted position wrong, so new observations cannot be linked to the right catalogue object;
3. **sensor scheduling**, because the uncertainty of an unpredicted manoeuvre has to be shrunk by tasking extra observations.

Methods to detect *past* manoeuvres exist; the gap the brief identifies is predicting *future* manoeuvre patterns by **reconstructing the analytical control law from time series of measurements**.

The project in one sentence

Simulate a station-keeping satellite with realistic physics, then show — step by step and with honest error bars — that its burns can be detected from noisy data, that the control law behind them (“burn when the mean SMA reaches the lower edge, stop at the upper edge”) can be learned, and that the learned law forecasts future burns well enough to help screening, association and tasking.

1.1 How the work progressed

Weeks 2–4 Basic simulation (gravity + drag + J_2), an *impulsive* deadband controller, inverse fitting of the drag coefficient, an FFT test for the J_2 signature, simple burn detectors (threshold, CUSUM, k -means, GMM), a held-out comparison and a neural-network extrapolation test.

Week 5 A *smooth* controller (tanh switches, a 7th state for the thruster, the mean SMA including the J_2 potential), burn detection in “ $\ddot{r}$ / $\dot{v}$ space” with a GMM, and event-level scoring.

Report steps 1–9 A complete, documented pipeline on 400-day simulations: dynamics, regression, drag and noise, spectra and Kalman filtering, deadband control, classification, robustness, repetition (stacking and Gaussian processes) and SINDy/BINDy control-law learning with forecasts.

Step 0 and audits Bias–variance tuning of every adjustable setting, walk-forward / walk-backward choice of the derivative method, a common figure style, and an audit of all figures.

Steps 10–18 Extensions towards the brief, using a fast mean-element model calibrated on the full simulation.

Step 19 Learning the thrust law and the natural dynamics together or apart, compared with the true model, and learning the natural law with no deadband.

All the old workstream stages were later re-written to use the *same* 400-day data as the report, so every figure in `outputs/report/` comes from one consistent data set.

2 Units, reference orbit and numerical integration

2.1 Non-dimensional units

The simulation is non-dimensional: the length unit is the Earth radius and the time unit is the inverse Earth spin rate,

$$\text{DU} = 6371 \text{ km}, \quad \text{TU} = 1/\omega_{\oplus} = 13\,713.44 \text{ s}, \quad \mu = \mu_{\oplus} \text{TU}^2/\text{DU}^3 = 289.873. \quad (1)$$

Working in these units keeps all state components of order one, which helps the ODE solvers and the least-squares fits (Section 6 shows why badly scaled columns matter).

2.2 Reference orbit

A Starlink-like circular orbit at 550 km altitude (SMA ≈ 6922 km), inclination 53° , period $T \approx 95.5$ min. The station-keeping band is $[6921.9, 6922.4]$ km in *mean* SMA (500 m wide), the same as Figure 1 of the brief.

2.3 Integrators

The uncontrolled runs use DOP853 (8th-order Runge–Kutta) with relative and absolute tolerance 10^{-12} (non-dimensional). The controlled runs use LSODA, which switches automatically between a non-stiff Adams method and a stiff BDF method: the tanh switches of the controller make the equations stiff for a few seconds around every burn edge and non-stiff in between. A solver cross-check in step 1 confirms that different integrators agree.

3 The dynamical model (report step 1)

3.1 Equations of motion

The state is $\mathbf{x} = (\mathbf{r}, \mathbf{v})$ (plus the thruster state s for the controlled runs). The acceleration is the sum of the forces:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \underbrace{-\frac{\mu \mathbf{r}}{r^3}}_{\text{two-body}} + \mathbf{a}_{J_2} + \mathbf{a}_{\text{drag}} + \mathbf{a}_{\text{Moon}} + \mathbf{a}_{\text{Sun}} + \mathbf{a}_{\text{SRP}} + \mathbf{a}_{\text{thrust}}. \quad (2)$$

Earth oblateness (J_2). The dominant perturbation at 550 km comes from the equatorial bulge. With R_e the **equatorial** radius (6378.137 km) and $J_2 = 1.08263 \times 10^{-3}$,

$$\mathbf{a}_{J_2} = \frac{3}{2} \frac{J_2 \mu R_e^2}{r^4} \begin{pmatrix} \frac{x}{r} \left(5 \frac{z^2}{r^2} - 1 \right) \\ \frac{y}{r} \left(5 \frac{z^2}{r^2} - 1 \right) \\ \frac{z}{r} \left(5 \frac{z^2}{r^2} - 3 \right) \end{pmatrix}, \quad U_{J_2} = \frac{\mu J_2 R_e^2}{2r^3} (3 \sin^2 \phi - 1), \quad (3)$$

where ϕ is the latitude; $\mathbf{a}_{J_2} = -\nabla U_{J_2}$ (checked by a unit test).

A bug found and fixed: the J_2 reference radius

The code originally used the *mean* Earth radius (6371 km) in the J_2 term. J_2 is defined with the *equatorial* radius, so the perturbation was $(6371/6378.137)^2 \approx 0.22\%$ too weak. This was fixed everywhere. The check (step 1): the theoretical nodal regression rate

$$\dot{\Omega} = -\frac{3}{2} n J_2 \left(\frac{R_e}{a} \right)^2 \cos i \quad (4)$$

integrated with the run's own (slowly drag-decaying) $a(t)$ agrees with the simulated right ascension of the ascending node to **0.04%** after 400 days ($-4.526^\circ/\text{day}$ at the start, $-4.642^\circ/\text{day}$ at the end, the change being due to drag lowering the orbit).

Atmospheric drag. With a constant (lumped) ballistic coefficient c_d ,

$$\mathbf{a}_{\text{drag}} = -c_d |\mathbf{v}| \mathbf{v}. \quad (5)$$

Two values exist in the code: the Week 2–4 demonstration value `CD_TRUE` (about 45 km/day of SMA decay, which would deorbit the satellite in ~ 12 days) and the *realistic* value `SMOOTH_CD` (≈ 0.13 km/day, i.e. 5.3 m/hour). Every 400-day run uses the realistic value. Steps 10–16 add a time-varying density (space weather).

Third bodies (Moon, Sun). The tidal (differential) attraction of a body at $\mathbf{r}_b$ with gravitational parameter μ_b is

$$\mathbf{a}_b = \mu_b \left(\frac{\mathbf{r}_b - \mathbf{r}}{|\mathbf{r}_b - \mathbf{r}|^3} - \frac{\mathbf{r}_b}{|\mathbf{r}_b|^3} \right), \quad (6)$$

the second term removing the attraction of the Earth itself (the origin is Earth-centred). Moon and Sun move on circular equatorial orbits (27.32 and 365.25 days).

Solar radiation pressure (SRP). A flat plate always facing the Sun, no eclipses: $|\mathbf{a}_{\text{SRP}}| = P C_R A/m$ with $P = 4.56 \mu\text{N}/\text{m}^2$, $C_R = 1.3$, $A/m = 5/260 \text{ m}^2/\text{kg}$, directed away from the Sun.

3.2 The force budget

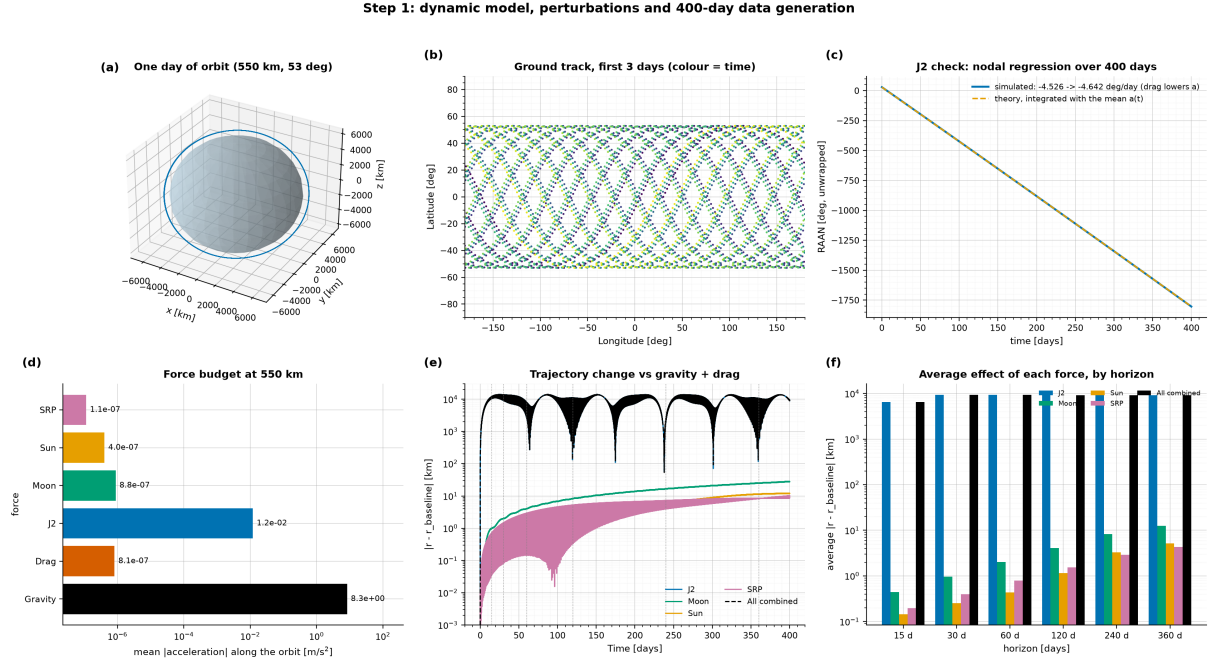


Figure 1: Step 1 summary: orbit, ground tracks, the J_2 nodal regression check, the force budget, and how far each force moves the satellite over 15–360 days.

At 550 km the mean accelerations are: gravity 8.3 m/s^2 , J_2 1.2×10^{-2} , Moon 8.8×10^{-7} , drag 8.1×10^{-7} , Sun 4.0×10^{-7} and SRP $1.1 \times 10^{-7} \text{ m/s}^2$. J_2 is four orders of magnitude above everything else, and the thruster (about $2 \times 10^{-4} \text{ m/s}^2$) sits between J_2 and drag. Over a year the J_2 -free orbit drifts by the orbit's diameter ($\sim 10^4 \text{ km}$) relative to the true one, while the Moon, Sun and SRP each move it by 10–30 km.

3.3 The 400-day data sets

All later steps read the same cached runs (`deadband/long_run.py`, about 500 MB in `.cache/`):

Run	Content	Used by
natural run	all forces, realistic drag, no thruster, 1-min samples	steps 1–4
divergence runs	gravity + drag, plus each extra force on its own	step 1
controlled run	smooth deadband controller, J_2 + drag, 6-s samples (102 burns)	steps 5–9
drag-only run	the controlled orbit without a thruster	step 5

Steps 6–9 learn from days 0–40 (TRAIN) and are tested on days 40–400 (TEST); results are reported at horizons of 15, 30, 60, 120, 240 and 360 days.

4 The station-keeping controller (report step 5)

4.1 Why the Week 4 controller had to change

The Week 4 controller fired an *impulsive* raise burn whenever the *osculating* SMA fell below the lower edge, plus a trim burn. Two problems:

1. **J_2 breaks the trigger.** The osculating SMA computed from the two-body energy $a_{\text{osc}} = -\mu/(2\varepsilon)$, $\varepsilon = v^2/2 - \mu/r$, swings by about 12 km per orbit under J_2 (twice per orbit: high over the equator, low at the highest latitudes). A 500 m band is therefore crossed every orbit: a controller watching the osculating SMA fires constantly.
2. **“If” statements are not ODEs.** An instantaneous impulse is a discontinuity that the solver handles only with event detection, and it does not represent a real (finite-thrust) burn.

4.2 The mean SMA

Including the J_2 potential in the energy removes the swing:

$$\varepsilon_{J_2} = \frac{v^2}{2} - \frac{\mu}{r} + U_{J_2}(\mathbf{r}), \quad a_{\text{mean}} = -\frac{\mu}{2\varepsilon_{J_2}}. \quad (7)$$

Since J_2 is conservative, ε_{J_2} is constant under J_2 and changes only through drag and thrust. To first order, $a_{\text{osc}} \approx a_{\text{mean}} - (J_2 R_e^2/a)(3\sin^2\phi - 1)$: the swing is +6.4 km over the equator and −5.8 km at 53° latitude. The simulated swing matches this to within 11 m; the orbit-averaged osculating SMA sits a fixed $(J_2 R_e^2/a)(1 - \frac{3}{2}\sin^2 i) = 275$ m above a_{mean} (a definition, not an error).

4.3 The smooth deadband law with memory

The state gains a 7th component, the thruster on/off state $s \in [0, 1]$. With the smooth switch $\sigma(x) = \frac{1}{2}(1 + \tanh(x/w))$ (width $w = 5$ m of SMA),

$$\tau \dot{s} = \underbrace{\sigma(a_L - a_{\text{mean}})(1 - s)}_{\text{switch on below the lower edge}} - \underbrace{\sigma(a_{\text{mean}} - a_U)s}_{\text{switch off above the upper edge}} + \text{latch}(s), \quad (8)$$

with a switch time $\tau = 60$ s and a weak latch term that pulls s towards 0 or 1, so the state *remembers* whether it is burning: between the edges both switches are off and s stays where it is. This is a **hysteresis**: the same SMA (say mid-band) can be coasting or burning depending on the past, which no memory-less function $s = f(a)$ can represent (step 9 measures exactly this). The thrust is

$$\mathbf{a}_{\text{thrust}} = T \theta(s) \hat{\mathbf{v}}, \quad \theta(s) = \sigma\left(\frac{s - 0.9}{0.02}\right), \quad T = 2 \times 10^{-4} \text{ m/s}^2, \quad (9)$$

so the throttle opens only once s is past 0.9 (a Hall-effect-class thruster, ~ 50 mN on ~ 250 kg).

Step 5: deadband control with a smooth (tanh) thruster, 10 and 400 days

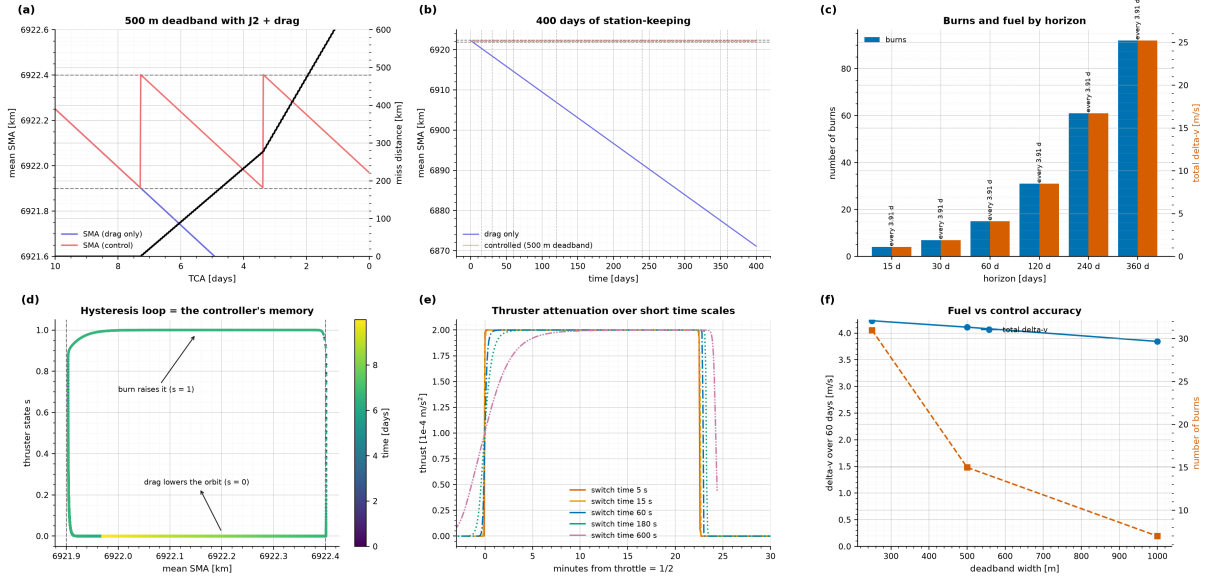


Figure 2: Step 5 summary: the reproduction of the brief’s Figure 1 (mean SMA, controlled vs drag-only, and the miss distance), 400 days of station keeping, burns and fuel by horizon, the hysteresis loop, the thruster ramp for several switch times, and fuel vs band width.

4.4 Results over 400 days

- 102 burns, one every **3.905 days**, each **22.8 min** long, $\Delta v = 0.273$ m/s per burn (27.8 m/s in total, 25.2 m/s over 360 days).
- The mean SMA stays in [6921.904, 6922.402] km for the whole run.
- The distance between the controlled orbit and its drag-only twin reaches 1800 km after 15 days and saturates near the orbit diameter, because the two satellites drift to different phases of the same orbit (the brief’s “hundreds of km in a week”).
- Fuel hardly depends on the band width (3.8–4.2 m/s over 60 days for 250–1000 m bands): the fuel replaces what drag removes; a narrower band only means more, smaller burns.

5 Step 0: bias–variance tuning of every setting

Many methods below have a free setting (a stencil order, a smoothing window, a number of mixture components, a sparsity threshold). Instead of picking them by hand, step 0 chooses each from data and saves the choice in `.cache/tuned_settings_v1.json`; every later step reads it with `settings.tuned(...)`, so the optimum is carried forward automatically.

5.1 The decomposition

Repeat an estimate $\hat{y}$ of a true value y over many independent noise draws (Monte-Carlo seeds or bootstrap re-fits):

$$\underbrace{\mathbb{E}[(\hat{y} - y)^2]}_{\text{MSE}} = \underbrace{(\mathbb{E}[\hat{y}] - y)^2}_{\text{bias}^2} + \underbrace{\mathbb{E}[(\hat{y} - \mathbb{E}[\hat{y}])^2]}_{\text{variance}}. \quad (10)$$

More flexible / less smoothed settings lower the bias and raise the variance; the best setting minimises the MSE. For time series the decomposition is done sample by sample and averaged.

5.2 The studies and what they chose

Setting	Default	Tuned	How
FD stencil order	6	per step: 1 min→4, 2 min→2, 5 and 10 min→6	MSE of c_d from 10 m-noise positions at each sampling step
Savitzky–Golay window / order	21 / 5	61 / 5	MSE of the classifier’s $\dot{v}$ (coast and burn weighted equally)
stacking window (step 8)	21 / 5	41 / 5	$\text{bias}^2 + (\text{noise})^2 \text{ variance}/N$: averaging N burns removes variance, not bias
derivative method	SG	SG	walk-forward / walk-backward event F1 (below)
GMM components K	4	4	$\text{bias}^2 + \text{variance of } P(\text{burn} \mid x)$ (Brier score), one-standard-error rule
SMA derivative window	41	41	MSE of the learned coast and burn rates
STLSQ threshold	0.05	0.05	MSE of the SINDy law’s predicted $\dot{a}$ on TEST (sparsest within 1%)

Why this way? A single global stencil order was first chosen on 1-minute data; the figure audit then showed that the 10-minute Kalman comparison of step 4 had silently inherited it and become 10% wrong. Truncation error grows like h^p while noise amplification falls with h , so the best order depends on the step h — hence an order *per sampling step*. The same logic gave step 8 its own (shorter) window: when 92 burns are averaged the variance falls by 92 but the bias does not.

5.3 Choosing the derivative by walk-forward and walk-backward testing

For each way of differentiating the velocity data (Savitzky–Golay, central differences of order 2/4/6/8), a GMM is trained on a 40-day window and tested out of sample over the following (*forward*: train days 0–40) or preceding (*backward*: train days 360–400) 15, 30, 60, 120, 240 and 360 days, at 0.25, 0.5 and $1\times$ the reference noise. At the reference noise only Savitzky–Golay keeps an event F1 of 1.0 in all 12 periods (0 false events, 27s burn-start error); the central differences fail from $0.5\times$ noise (8th order already at $0.25\times$) because they amplify noise.

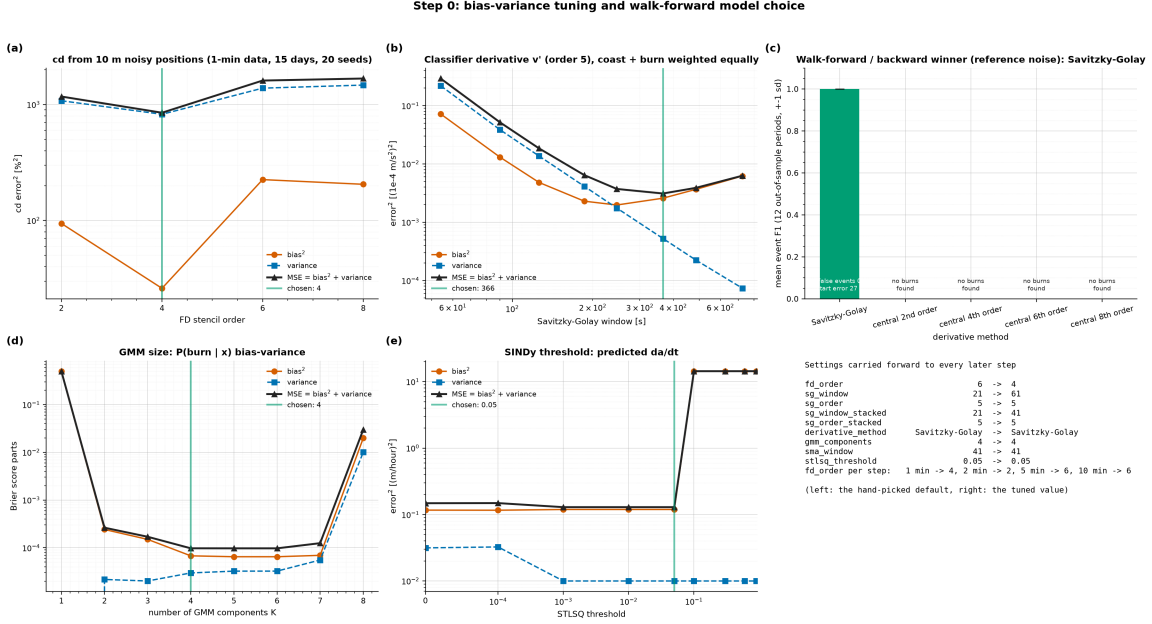


Figure 3: Step 0 summary: bias², variance and MSE for each setting, and the walk-forward / walk-backward winner.

6 Step 2: learning the coefficients by regression

6.1 Finite differences

From positions $\mathbf{r}_k$ sampled every h , central differences of order p give the velocity and acceleration, e.g. for $p = 2$:

$$\dot{\mathbf{r}}_k \approx \frac{\mathbf{r}_{k+1} - \mathbf{r}_{k-1}}{2h}, \quad \ddot{\mathbf{r}}_k \approx \frac{\mathbf{r}_{k+1} - 2\mathbf{r}_k + \mathbf{r}_{k-1}}{h^2}, \quad (11)$$

and wider stencils for $p = 4, 6, 8$. The truncation error is $O(h^p)$; white position noise σ becomes an acceleration error of order σ/h^2 (amplified by the stencil weights). This is the central bias–variance trade-off of the whole project.

6.2 Linear least squares

Each unknown multiplies a known “shape”, so one linear solve recovers it:

$$\ddot{\mathbf{r}} = \mu \left(-\frac{\mathbf{r}}{r^3} \right) \quad \Rightarrow \quad \hat{\mu} = (\Theta^\top \Theta)^{-1} \Theta^\top \ddot{\mathbf{r}}, \quad (12)$$

$$\ddot{\mathbf{r}} + \frac{\mu \mathbf{r}}{r^3} - \mathbf{a}_{J_2} = c_d (-|\mathbf{v}|\mathbf{v}), \quad (13)$$

$$\ddot{\mathbf{r}} + \frac{\mu \mathbf{r}}{r^3} = [c_d, J_2, \mu_{\text{Moon}}, \mu_{\text{Sun}}, PCA/m] \cdot [\text{shapes}]. \quad (14)$$

The columns differ by up to 10 orders of magnitude (the Sun term is tiny), so they are scaled to unit length before solving and the scaling is undone afterwards (same answer, no round-off trouble).

Step 2: learning the coefficients by regression (400-day data)

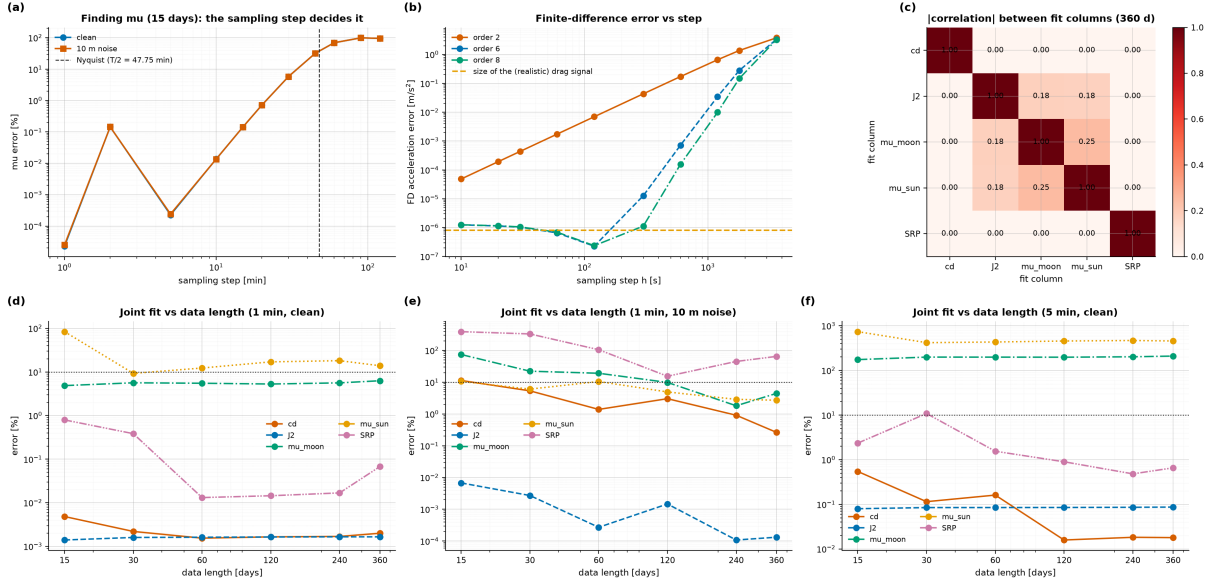


Figure 4: Step 2 summary: μ vs sampling step (the Nyquist limit $T/2 = 47.75$ min), finite-difference error vs step and order, the correlation of the joint-fit columns, and the joint fit vs data length (clean 1-min, 10 m noise, clean 5-min).

Results. On clean 1-min data the joint fit recovers c_d and J_2 to $\sim 0.002\%$. The Moon ($\sim 5\%$) and especially the Sun ($10\text{--}80\%$) are poorly identifiable: their tidal shapes are small and correlated with each other and with J_2 (column correlations $0.18\text{--}0.25$). With 10 m noise the errors fall with data length (e.g. c_d from 11% at 15 days to 0.27% at 360 days). On 5-minute data the truncation error swamps the small terms (Moon and Sun $200\text{--}700\%$ off).

7 Step 3: drag estimation and noise robustness

7.1 Three ways to get c_d

1. **FD regression** (above): two derivatives of noisy positions.
2. **The energy method**: one derivative only. The mean SMA from the J_2 energy decays at

$$\dot{a} = -\frac{2a^2}{\mu} c_d |\mathbf{v}|^3, \quad (15)$$

so c_d follows from the *slope* of a straight line fitted through the whole SMA history: $\hat{c}_d = -\dot{a} \mu / (2\bar{a}^2 |\mathbf{v}|^3)$. Fitting a line averages the noise of every sample.

3. **Shooting**: propagate the orbit with trial parameters and fit the whole trajectory to the observed positions by Levenberg–Marquardt (started from the energy-method answer, because the cost is rugged far from the minimum).

noise (15 days, 1-min)	0 m	1 m	5 m	10 m	50 m	100 m
FD regression c_d error [%]	0.009	0.64	3.2	6.3	32	64
energy method c_d error [%]	0.006	0.008	0.015	0.035	0.18	0.35

With 10 m noise the energy method improves from 0.03% (15 days) to 0.001% (360 days); shooting gives $0.05\% / 0.04\%$ at 15 / 30 days at a much higher cost.

Step 3: drag estimation, noise robustness and data length

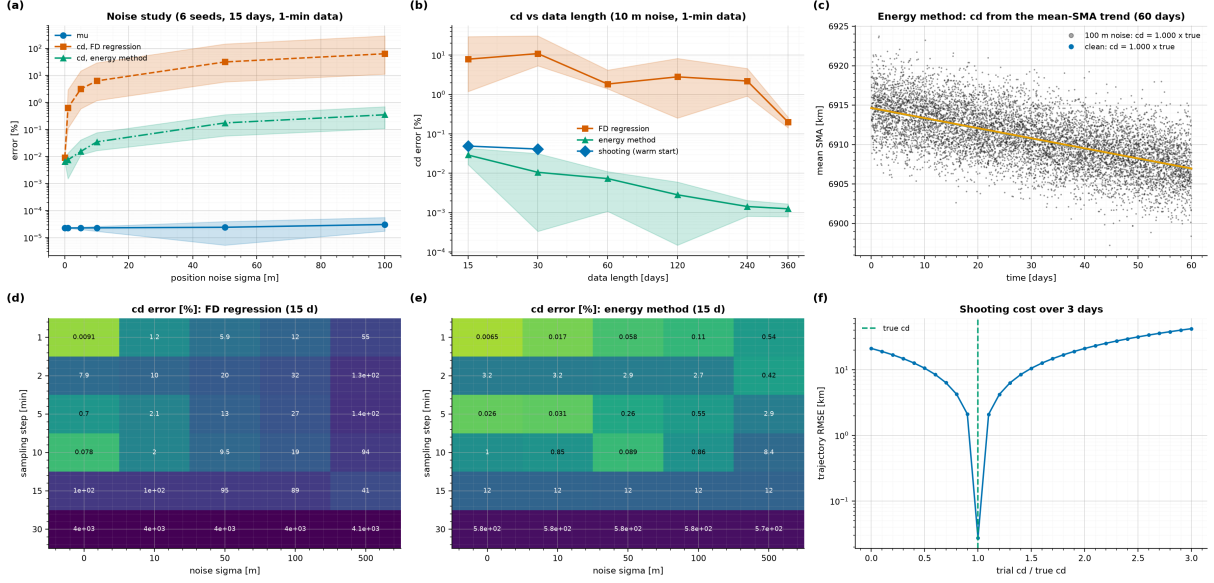


Figure 5: Step 3 summary: noise study, c_d vs data length, the energy method’s SMA trend, sensitivity maps (sampling step $\times$ noise) and the shooting cost landscape.

7.2 Step 3b: realistic (coloured) noise

Real tracking errors are not white. The realistic mixture combines white noise at four rates (every sample and held for 10/100/1000 samples), pink ($1/f$) noise, a random walk ($1/f^2$) and once-per-orbit and once-per-day sinusoids, scaled to the *same* RMS as the white case so that only the spectrum differs. Second derivatives multiply the noise spectrum by f^4 (fast noise hurts, slow noise is filtered); the energy method fits a slow trend, so slow noise (drifts, random walk) can mimic drag decay. At 1 m RMS over 30 days the realistic noise makes the energy-method c_d $4.2\times$ worse, SRP $4.0\times$ and the Sun $2.3\times$ worse, while FD c_d and J_2 are almost unchanged.

7.3 Workstream E: whole-trajectory cost surface

The shooting fit of (c_d, J_2) on 3 days of 10-min data has a cost surface (`inverse_cost_surface_3d`) with a narrow valley at the true c_d running along J_2 ; the joint fit at 100 m noise recovers c_d to 0.006% and J_2 to 0.0014%, with a 27 m RMS floor that is model error (the observed orbit also feels the Moon, Sun and SRP, which the fitted model omits).

8 Step 4: spectra, filtering and Kalman filtering

8.1 The periodogram and the J_2 signature

The radial unmodelled acceleration (finite-difference acceleration minus point-mass gravity and drag) is Fourier-transformed with a Hann window. J_2 depends on $\sin^2 \phi$, so it appears at **twice the orbital frequency**. The signal-to-noise ratio is the peak power within $\pm 1\%$ of 2 cycles/orbit divided by the median background power below 6 cycles/orbit (a fine sampling step must not be penalised by high-frequency finite-difference noise far above the band of interest).

- SNR falls with noise (15 days, 5-min: 47 dB at 0.1 km, 27 dB at 1 km, 12 dB at 5 km).
- SNR collapses once the sampling step exceeds the J_2 Nyquist limit $T/4 \approx 23.9$ min (55 dB at 1 min, -3 dB at 90 min, step 4c).

- A longer record gives finer frequency resolution, but over months drag slowly shifts the line, so SNR peaks at ~ 60 days.

8.2 Filtering

A band-pass around 2 cycles/orbit isolates J_2 and a low-pass below 0.3 cycles/orbit isolates drag. For *estimating* a force with a known shape, filtering does not beat least squares, which already is the matched filter.

8.3 The extended Kalman filter (EKF)

The state $\mathbf{x} = (\mathbf{r}, \mathbf{v}, c_d)$ is propagated with the full gravity + J_2 + drag model and its state-transition matrix Φ ($\dot{\Phi} = F\Phi$, F the Jacobian); each noisy position updates it:

$$\text{predict: } \mathbf{x}^- = f(\mathbf{x}), \quad P^- = \Phi P \Phi^\top + Q, \quad (16)$$

$$\text{update: } K = P^- H^\top (H P^- H^\top + R)^{-1}, \quad \mathbf{x} = \mathbf{x}^- + K(\mathbf{z} - H\mathbf{x}^-), \quad P = (I - KH)P^-. \quad (17)$$

With 10-min data and 10 m noise the EKF finds c_d to 0.04% after 15 days and 0.01% after 30 days, better than the energy method (1.2%) and FD regression (0.5–21%) on the same coarse data.

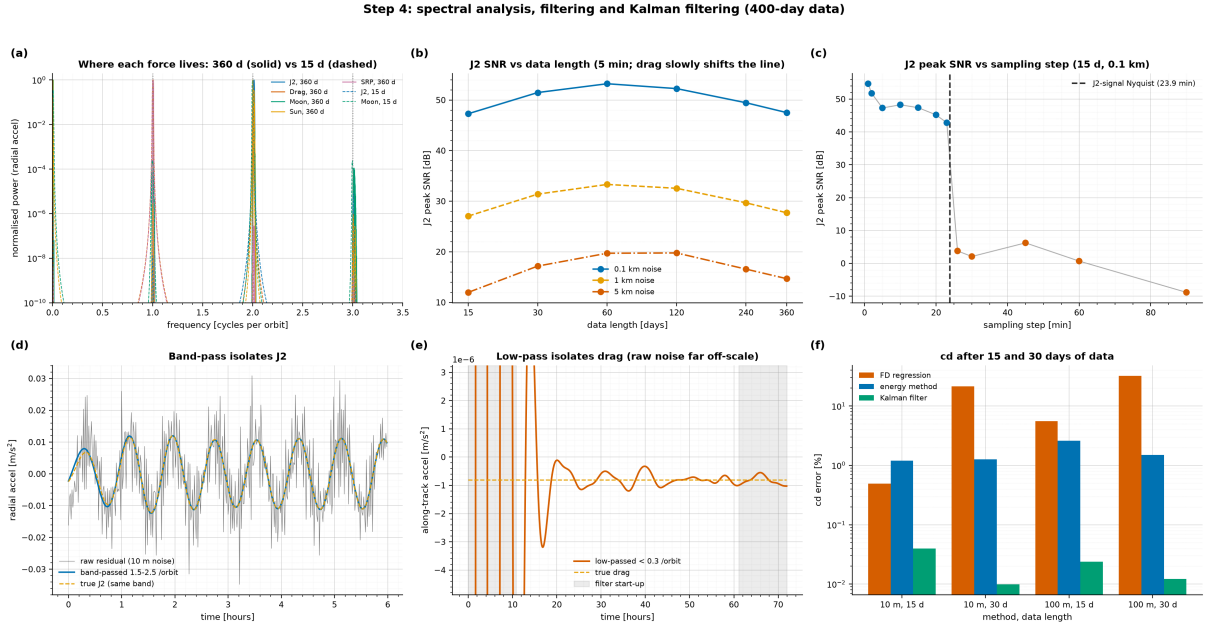


Figure 6: Step 4 summary: where each force lives in frequency, J_2 SNR vs data length and sampling step, band-pass and low-pass filtering, and the Kalman filter against the other c_d methods.

8.4 Step 4b: “force spectroscopy”

Inspired by infrared spectroscopy, each force is identified by its *diagnostic band* (a band where it holds $\geq 90\%$ of the power) and its strength read from the band area relative to a reference library. Parseval’s theorem ($\sum |X_k|^2 \propto \overline{a^2}$) explains a key result: spectral *power* shares do not match the force budget (they square the amplitudes, so J_2 ’s 99.98% becomes 99.999999%); the *square root* of the power does (RMS $\approx$ mean $|a|$). Band reading is 30–40 $\times$ faster than regression; it recovers drag to 0.5%, J_2 to 2% and SRP to 0.9% in random mixtures, the Moon to 8%, while the Sun is not separable (its lines lie under the Moon’s). A phase-aware fit is exact but needs the force model evaluated along the orbit, like regression.

9 Step 6: classifying coast and burn

9.1 The feature: the unmodelled acceleration

From measured $\mathbf{r}$, $\mathbf{v}$ (6-s samples with reference noise 0.1 m and 0.5 mm/s), the unmodelled acceleration is the derivative minus the known physics:

$$\mathbf{a}_u = \dot{\mathbf{v}}_{\text{measured}} - \left(-\frac{\mu \mathbf{r}}{r^3} + \mathbf{a}_{J_2} \right) \approx \mathbf{a}_{\text{drag}} + \mathbf{a}_{\text{thrust}}. \quad (18)$$

While coasting $|\mathbf{a}_u| \sim 10^{-6} \text{ m/s}^2$ (drag); during a burn $\sim 2 \times 10^{-4}$. Two features are used: $x_1 = \log |\mathbf{a}_u|$ (how big) and $x_2 = \mathbf{a}_u \cdot \hat{\mathbf{v}}/10^{-4}$ (thrust pushes *forwards*, noise does not). Velocity-based ($\dot{v}$) features are used rather than position-based ($\ddot{r}$) ones because a second derivative of noisy positions is pure noise.

Why this way? Why the derivative method matters: with the 2nd-order stencil at 6 s the coast residual is the stencil’s own truncation error ($\sim 6 \times 10^{-5} \text{ m/s}^2$), only $4\times$ below the thrust, so the GMM’s burn Gaussian was not centred on the burns. Higher-order stencils bring the floor to the real drag (8×10^{-7}) on clean data; with noise only the Savitzky–Golay filter (a least-squares polynomial through a window) works. The tuned window is 61 samples (6 min).

9.2 The Gaussian mixture model

A GMM models the feature density as $p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$, fitted by expectation–maximisation on TRAIN. The posterior (“weight”) of a sample is $P(k | \mathbf{x}) = \pi_k \mathcal{N}_k / \sum_j \pi_j \mathcal{N}_j$. With $K = 4$ (tuned): one or two Gaussians for the coast, one for the burn plateau and a spare for the thruster ramps; a component is labelled *burn* if its mean $\log |a_u|$ lies more than 3 coast standard deviations above the coast mean (thrust only adds acceleration). Alternatives: a Bayesian GMM (Dirichlet prior switches off unused components), k -means, Isolation Forest and a MAD threshold.

9.3 From samples to events

A 23-min burn is ~ 230 samples. Flags are grouped into **events**: gaps of ≤ 2 samples are filled and runs shorter than 10 samples (1 min) are dropped. Scoring is per event: a true burn is found if a detected event overlaps it within ± 60 s; precision = real events / detected events, recall = burns found / burns, F1 their harmonic mean.

9.4 Results

On 92 TEST burns over 360 days: GMM, Bayesian GMM and the MAD threshold score F1 = 1.0 on noisy data (all 92 burns, 0 false events), k -means fails completely with noise (136 662 false events), Isolation Forest 0.85. The scores hold at every horizon from 15 to 360 days.

9.5 Stress test: how coarse can the sampling get?

At 0–4 $\times$ the reference noise and steps from 6 s to 60 min:

- the Savitzky–Golay derivative works up to a **2-min** step (beyond, a polynomial through 5 samples spanning a large arc of the curved orbit has a truncation error bigger than the thrust);
- a new **one-step-propagation** feature — propagate each measured state one step with gravity + J_2 (RK4) and divide the velocity mismatch at the next measurement by the step — works from ~ 1 min up to **20 min** at every noise level, since its noise shrinks as $1/h$;

- switching between the two at about 2 min gives $F1 \geq 0.9$ from 6 s to 20 min at all noise levels; beyond one burn length (23 min) the burn is diluted into one interval.

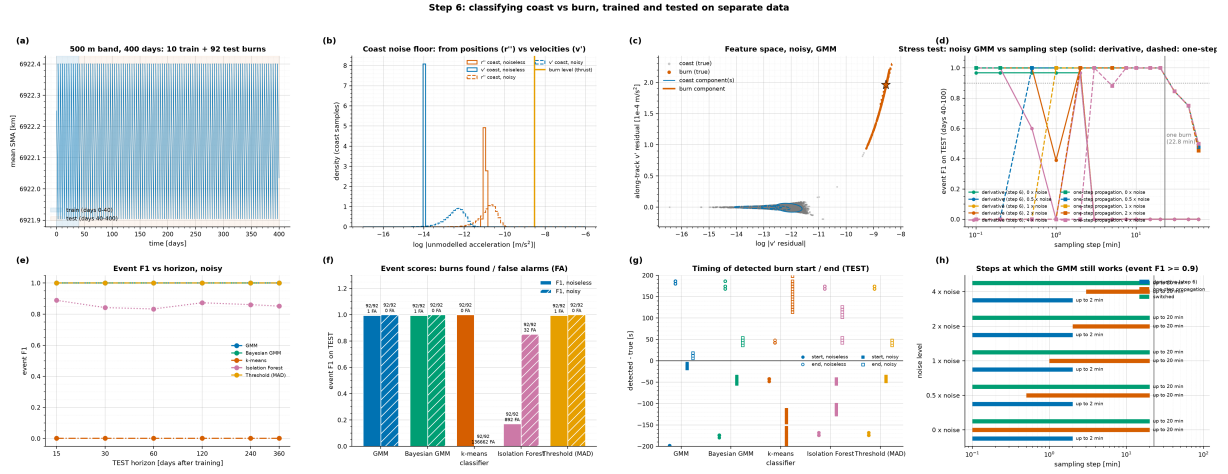


Figure 7: Step 6 summary: dataset split, $\ddot{r}$ vs $\dot{v}$ noise floors, the feature space with the GMM, the stress test, event F1 vs horizon, event scores, timing errors and the working ranges of the sampling step.

Phase space, days 40-48 (TEST), coloured by the GMM weight $P(\text{burn})$: the coast line (dark) and the burn line (bright) are told apart sample by sample

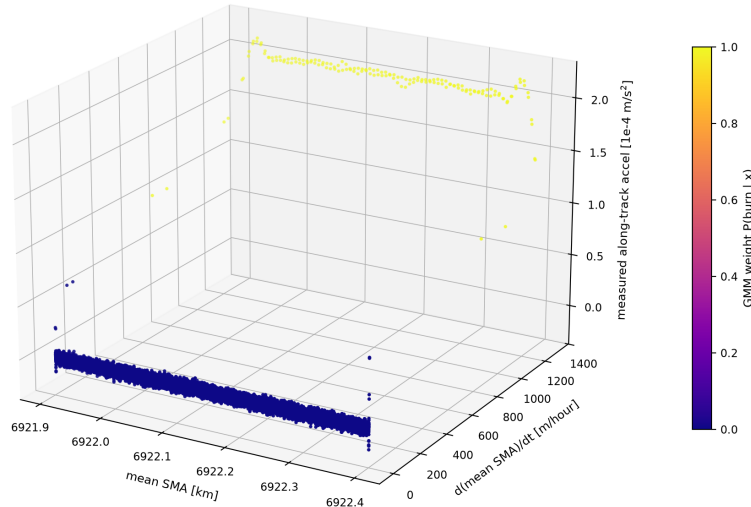


Figure 8: The phase space (mean SMA, its rate, along-track acceleration) over two TEST cycles, every sample coloured by its GMM weight $P(\text{burn} | x)$.

10 Step 7: robustness of the classifiers

Days 0–40 train, the next 60 days test. GMM, Bayesian GMM and threshold keep $F1 = 1.0$ from 0.25 to $8\times$ the reference noise; k -means fails as soon as there is noise (its two clusters split the coast cloud); Isolation Forest stays near 0.8–0.9. The GMM needs ~ 1000 training samples. The number of components hardly matters between 2 and 6 at these noise levels. The workstream **detection**

stage adds five basic detectors (rolling z-score, MAD threshold, k -means, GMM, CUSUM) on the same data: CUSUM is the most noise-tolerant online detector.

11 Step 8: repetition — autocorrelation, stacking and a Gaussian process

- The **autocorrelation** of the noisy along-track acceleration peaks at 3.9051 days, matching the true burn period (3.9051 days).
- **Stacking**: aligning the 92 detected burns and averaging them reduces noise; the error of the average falls until it hits a floor set by the smoothing of the edges (the tuned stacking window trades this bias against the $1/\sqrt{N}$ variance).
- A **Gaussian process** (RBF + white-noise kernel) through the stacked samples gives a thrust law: level $2.00 \times 10^{-4} \text{ m/s}^2$ (0.999 of the truth), duration 1368 s (exact), rise time 60 s (truth 30 s: the derivative smooths the edge).
- With only 15 days (4 burns) the level is already within 0.3%, the period exact to 10^{-4} .

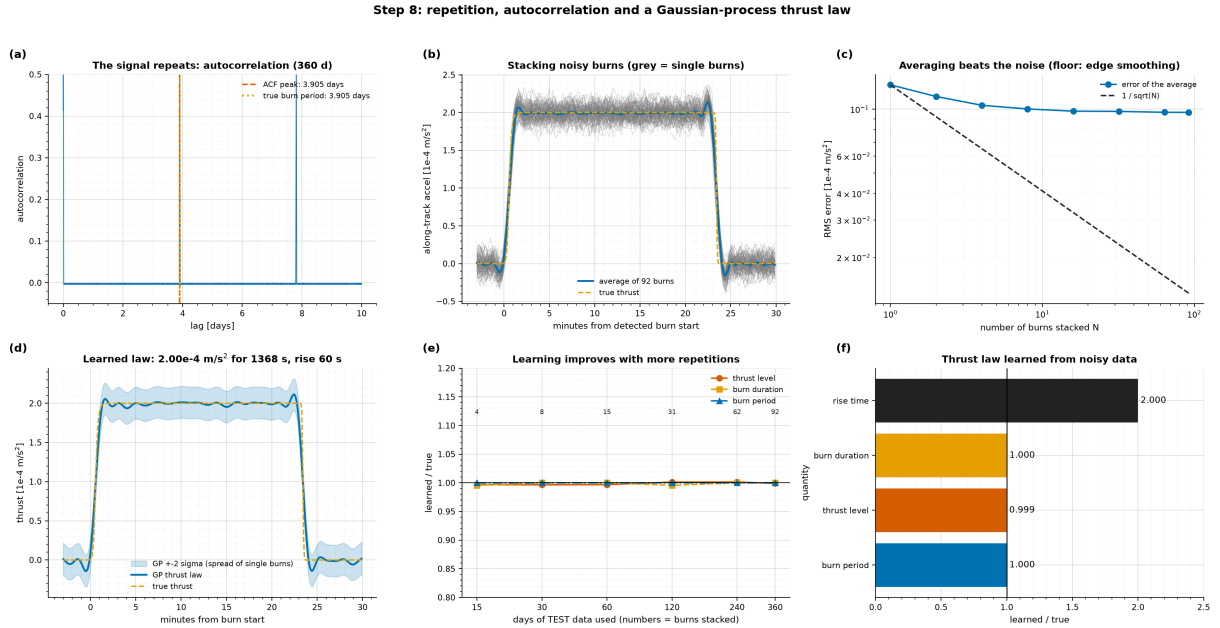


Figure 9: Step 8 summary.

12 Step 9: learning the control law (SINDy and BINDy) and forecasting

12.1 Sparse identification of nonlinear dynamics (SINDy)

SINDy writes the rate of a state as a sparse combination of candidate functions (a “library”):

$$\dot{a} = \Theta(a, s) \xi, \quad \Theta = [\theta_1(a, s), \theta_2(a, s), \dots], \quad (19)$$

and finds a sparse ξ by **sequentially thresholded least squares** (STLSQ): solve least squares, set coefficients below a threshold to zero, refit on the remaining terms, repeat.

Why a cluster-indicator library. The obvious physics library $[D(a), B(a), 1]$ with $D = \frac{2a^2}{\mu}|v|^3$ (drag shape) and $B = \frac{2a^2}{\mu}|v|$ (thrust shape) is singular: over a 500 m band both are constant to a few ppm and perfectly correlated (correlation 0.997). The library therefore uses the thruster state from the classifier: $[1 - s, s, a - a_{\text{ref}}, s(a - a_{\text{ref}}), (a - a_{\text{ref}})^2]$. STLSQ keeps only the two cluster terms; drag and thrust follow from them: $\hat{c}_d = 1.0005 c_d$ and $\hat{T} = 0.9997 T$.

BINDy (Bayesian). Automatic relevance determination (ARD) regression gives a posterior mean and standard deviation for each coefficient, i.e. credible intervals for the physics.

12.2 The switching law

The burn start and end SMAs give the learned edges 6921.9037 / 6922.4013 km. A memory-less sigmoid $s = f(a)$ reaches only 0.50 balanced accuracy (chance); the hysteresis with the learned edges reaches 1.00. **The controller has memory, and the learned law must too.**

12.3 Forecasting

The learned hybrid model — $\dot{a}$ from SINDy, switching by the learned hysteresis — is integrated forward from day 40 with no further data. Mean onset error over the 360-day forecast: periodic baseline 0.9 min, GP coast curve 18 min, BINDy ensemble 34 min, SINDy 195 min, random forest 286 min.

Why this way? Why does SINDy drift and why does the periodic baseline win here? A coast rate learned from 40 days of noisy $\dot{a}$ carries a few hundred ppm of error; each 3.9-day coast is then a little too short, and the error accumulates linearly over 92 cycles. The periodic baseline learns the period directly, which is perfect *only because drag is constant in this simulation*. Step 10 removes this assumption and the ranking reverses.

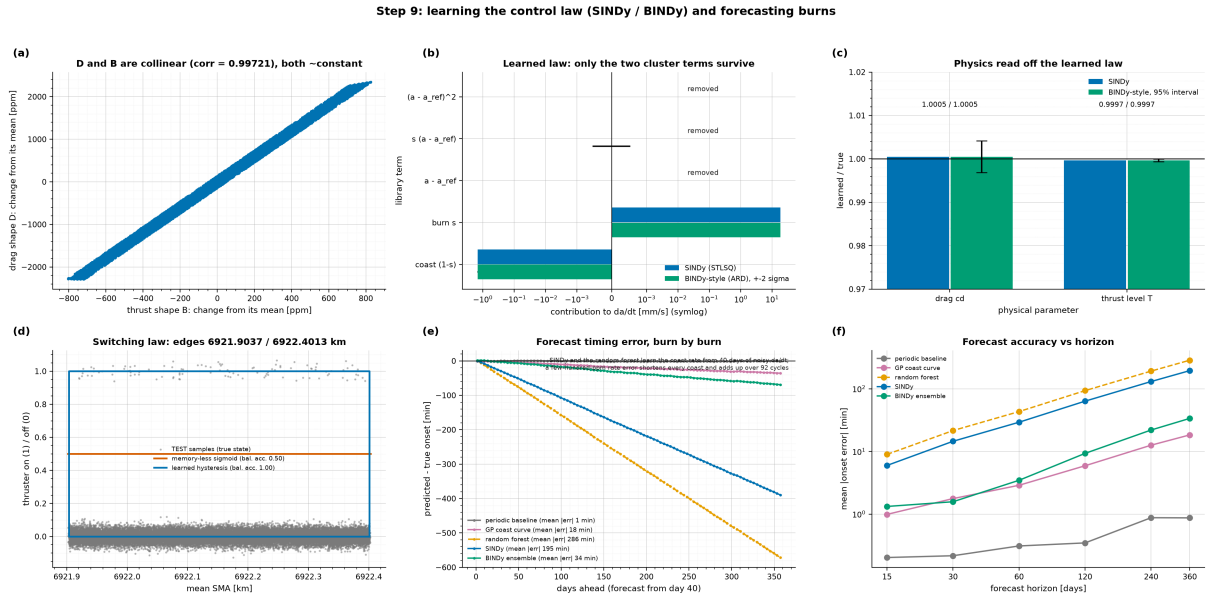


Figure 10: Step 9 summary.

13 The workstream stages, aligned with the report data

The Week 2–5 stages were rewritten to use the report’s data, split, features and event scoring, and gained 3D views:

neural_net an MLP fitted as $\text{SMA} = f(t)$ on days 0–40 flattens outside its training window (44 km off after 360 days) whereas a straight line is 0.09 km off: a time-indexed network cannot extrapolate physics.

inverse the 3D cost surface over (c_d, J_2) (above).

spectral the J_2 SNR surface over sampling step $\times$ noise and a waterfall of the spectrum against record length.

smooth_deadband, deadband Figure 1, the switches, the J_2 mean-vs-osculating figure, the 400-day sawtooth and a 3D hysteresis loop.

derivative_gmm the GMM in $\ddot{r}/\dot{v}$ space, five derivative methods, the Week 4 features on the same samples, and a cadence sweep.

gmm the GMM density and burn-probability surfaces in 3D, the phase space coloured by GMM weight, and GMM vs k -means decision regions.

heldout seven detectors on held-out days 40–100, including a Gaussian-process anomaly detector that models the coast SMA as a function of the time since the SMA left the upper edge (trained on GMM labels, not truth): GMM, Bayesian GMM, DBSCAN and the GP reach $F1 = 1.0$, Isolation Forest 0.71, k -means and Ward clustering 0.

14 Extensions towards the brief (steps 10–18)

14.1 The mean-element model

A full 6-s integration of 400 days takes about 20 minutes; the extensions need hundreds of scenarios. They use the model the controller itself follows:

$$\dot{a} = -k w(t) e^{-(a-a_{\text{ref}})/H} + T s, \quad s : 0 \rightarrow 1 \text{ at } a \leq a_L, 1 \rightarrow 0 \text{ at } a \geq a_U, \quad (20)$$

with k the coast decay rate (5.335 m/h), T the thrust contribution (1308 m/h), $H = 60$ km the density scale height and $w(t)$ the space-weather density factor. k , T and the *effective* edges (the tanh switch starts each burn ~ 6 m above the nominal lower edge) are calibrated on the full run; the model reproduces the burn period to 0.01% and the burn length to 1%. The effective band is the SMA span covered by a coast (coast rate $\times$ coast duration), about 3 m wider than the SMA difference between the throttle half-way points because of the thrust ramps.

14.2 Step 10: space-weather drag

$w(t)$ combines a 27-day solar-rotation wave, a slow solar-cycle trend and 8 geomagnetic storms (density $\times 2$ –4, 1.5-day decay); an observable daily proxy (an F10.7/Ap-like index) has 5% noise. Forecasters: periodic baseline; the law with constant drag; with the latest coast rate; with $k \times$ proxy and a 27-day persistence forecast of the proxy; and with a perfect space-weather forecast (upper bound). Forecasts are event-driven: with $G(t) = \int k \hat{w} dt$ a coast from a_0 ends where $G(t) - G(t_0) = a_0 - a_L$.

mean onset error [min]	const., 15 d	const., 360 d	space weather, 15 d	space weather, 120 d
periodic baseline	2	3	1642	1732
law, constant drag	68	630	1418	1670
law + space-weather proxy	68	630	730	1286
law + perfect forecast	68	630	164	419

Variable drag changes the interval between burns but not the band edges: the periodic baseline then loses its phase, while the learned law tracks it as well as drag can be forecast.

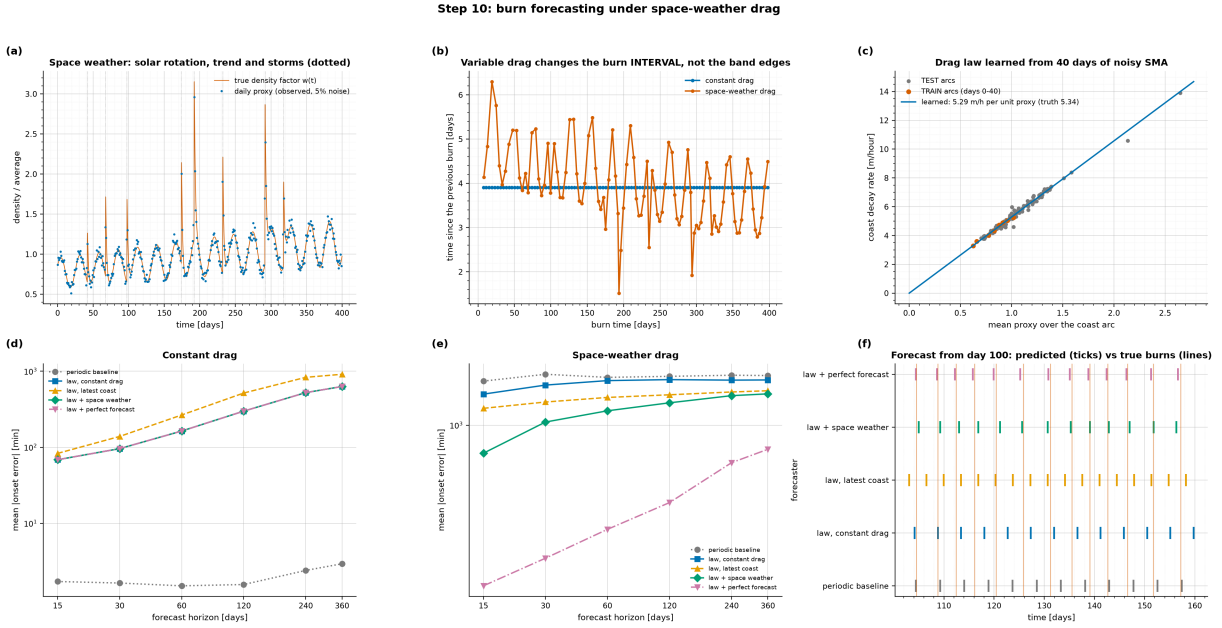


Figure 11: Step 10 summary.

14.3 Step 11: realistic tracking passes

Instead of 6-s states, 8-minute passes of 12-s positions (10 m–1 km noise), 1–8 per day. Each pass is fitted by batched Gauss–Newton orbit determination (6 state parameters, RK4 gravity + J_2 , finite-difference Jacobian); one pass gives the SMA to about $2\times$ the position noise. Burns are found in the gaps: the SMA jump minus the drag decay (estimated from pass pairs a day apart) must exceed a threshold set by the pass noise. Pass-to-pass: F1 ≈ 1 at 10 m noise for 1–8 passes/day; at 100 m only a multi-pass test (mean of all passes a day before vs after) finds some burns (F1 0.47 at 8/day); at ≥ 300 m the 500 m burn is buried. A CCSDS OEM reader runs the same analysis on real ephemerides placed in `data/ephemeris/`.

14.4 Step 12: close-approach screening

An SMA error δa changes the mean motion by $\delta n = -\frac{3}{2}(n/a)\delta a$, so the along-track error grows as

$$\delta s(t) = -\frac{3}{2}n \int_0^t (a_{\text{pred}} - a_{\text{true}}) dt; \quad (21)$$

an unknown 500 m burn makes the catalogue drift by $\frac{3}{2}n \times 0.5 \text{ km} \approx 3 \text{ km/hour}$. RMS along-track error at 1 / 7 days: drag-only 19 / 396 km, periodic 18 / 42 km, learned law with space weather 5.8 / 11.9 km, perfect forecast 2.0 / 4.1 km. Screening with $P_c \approx \pi R^2 \mathcal{N}(\text{miss}; 0, \Sigma)$ ($R = 20 \text{ m}$, alert at 10^{-4}): with an honest covariance even the best method misses 25% of dangerous conjunctions at 1 day (drag-only 68%), because the errors are still km-sized — the “dilution” that the brief

warns makes screening ineffective. With today’s catalogue-only covariance nearly all dangerous conjunctions are missed.

14.5 Step 13: sensor tasking and track association

Six observation opportunities a day, a budget of 1, 2 or 4. With an association gate of 20 km, custody is lost after 74% of burns with one random observation a day, 48% if the observation is scheduled just after each *predicted* burn, and 15–19% if the catalogue also carries the predicted burn (its error is then only $\frac{3}{2}n\delta a \times$ timing error). With 4 observations a day the losses are 20% and 4%.

14.6 Step 14: several control laws on one satellite

Orbit raising from 6890 km, the 500 m deadband, a 3 km band change at day 150 and six 3-min collision-avoidance burns. From noisy SMA (5 m, 10-min samples) every burn is detected and labelled from duration, SMA gain and start SMA relative to a running edge: **105/105 correct** by rules and also by an unsupervised GMM; the band change is found as a changepoint at day 150.0; each regime gets its own rate law.

14.7 Step 15: a fleet of 40 satellites

Forty satellites share the space weather but differ in band (± 5 km), drag ($\times 0.7$ – 1.3) and thrust ($\pm 10\%$). Three pattern-of-life changes are planted (band up 1 km at day 200, thrust halved at day 250, band narrowed to 200 m at day 300). Rolling 30-day laws are compared with each satellite’s own history, with the coast rate divided by the fleet median (removing the common space weather): exactly the three changed satellites are flagged, 10–20 days after each change, with no false flags. Empirical-Bayes pooling of the drag coefficient, $\hat{k}_i^{\text{pool}} = (\tau^2 \hat{k}_i + s_i^2 \bar{k})/(\tau^2 + s_i^2)$, reduces the error from 21.6% to 12.5% with 4 days and from 11.8% to 6.8% with 8 days of data per satellite (with only 2 days the single-arc variance has to be guessed and pooling overshoots).

14.8 Step 16: calibrated (conformal) burn-time intervals

Split conformal prediction: the absolute onset errors of forecasts from calibration origins (days 40–200) give, per lead-time bucket, the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile q ; the interval is $\pm q$. On new origins (days 200–360) the 90% intervals of the space-weather forecaster cover 93–99% of the burns.

14.9 Step 17: GEO station keeping

The GEO longitude obeys $\ddot{\lambda} = -A \sin 2(\lambda - \lambda_s)$ (Earth’s triaxiality, $A = 0.0017^\circ/\text{day}^2$, $\lambda_s = 75.07^\circ\text{E}$); the inclination grows by $0.85^\circ/\text{year}$. East-west (impulsive, $\pm 0.05^\circ$ box), north-south (impulsive) and an electric-propulsion east-west law (continuous low thrust with a deadband on position and drift rate) are simulated. SINDy recovers $A = 0.00171$ and $\lambda_s = 74.98^\circ$ from a free-drift phase; inside the box only the local acceleration is identifiable. Impulses are detected from 6-hourly longitudes as drift-rate jumps between 8-day quadratic fits, and the east-west burns are forecast 0.1–1.3 days off up to 225 days ahead from the learned local acceleration and box edges.

14.10 Step 18: the SINDy equation zoo

Ten candidate libraries per dataset — constant per cluster, polynomial, exponential density, time polynomial, Fourier in time, space-weather proxy, proxy $\times$ density, a deliberately wrong library

without a cluster indicator and a “kitchen sink”; for GEO polynomials and Fourier harmonics in λ :

- on the full-physics data every richer library is pruned back to the same two-term law (the data support nothing more);
- with space weather only the proxy libraries forecast well (about 260 vs ~ 1640 min);
- for GEO the true $\sin 2\lambda, \cos 2\lambda$ form (with or without a constant) forecasts best;
- “kitchen-sink” libraries diverge through collinearity; BIC alone can prefer an over-fitted library that forecasts worse, so free-running forecasts are the decisive test;
- **thresholding**: classic STLSQ drops terms small *relative to the largest* coefficient, which deletes the real coast drag next to the $250\times$ larger, rare burn term; a significance threshold ($|\xi| \geq 3$ standard errors) keeps it but lets spurious terms through on very large, correlated data sets. Both are reported side by side.

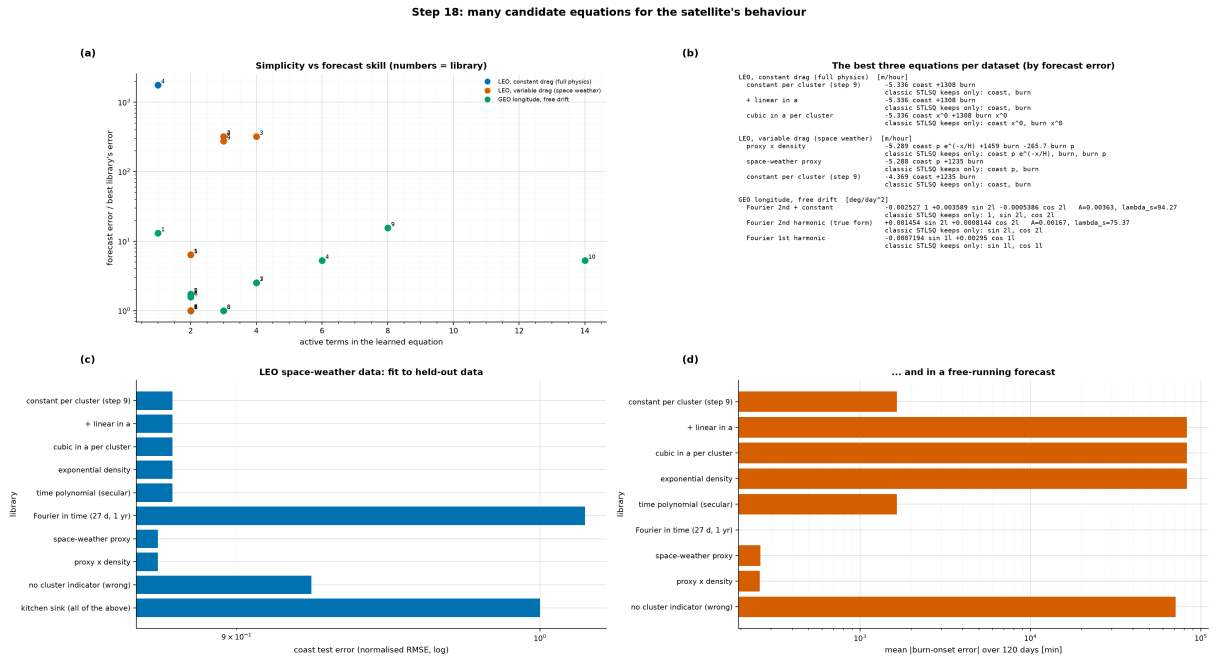


Figure 12: Step 18 summary.

15 Step 19: thrust law and natural dynamics, learned together or apart

15.1 Why split the problem?

Step 9 learned everything at once. But the mean-SMA dynamics are the sum of two physically different parts,

$$\dot{a} = \underbrace{-c_d D(a, |\mathbf{v}|)}_{\text{natural (coasting) dynamics}} + \underbrace{T \theta(t) B(a, |\mathbf{v}|)}_{\text{thrust control law}}, \quad D = \frac{2a^2}{\mu} |\mathbf{v}|^3, \quad B = \frac{2a^2}{\mu} |\mathbf{v}|, \quad (22)$$

plus the switching law that decides θ (on at the lower edge, off at the upper edge). In operations one part is often better known than the other: the natural dynamics come from a catalogue force model, while the operator's thruster and control law are unknown — or, for one's own satellite,

the thrust is known and the drag is not. Step 19 measures what each kind of prior knowledge is worth.

15.2 The four scenarios

All use step 9’s data (the 400-day controlled run, reference noise, smoothed derivative $\dot{a}$, burn flags from the GMM), learn on days 0–40 and are scored on days 40–400 against the *true* model (c_d , T , the true throttle profile and the true effective edges). Each is fitted by least squares (SINDy point estimate) and by ARD regression (BINDy posterior standard deviation).

- A. Whole system** one regression of $\dot{a}$ on $[-D, sB]$ over *all* samples, with s the detected burn flags; edges from the detected burns. Everything learned at once, as it comes.
- B. Thrust law only (natural known)** c_d is given: $\dot{a} + c_d D = T s B$ gives T ; edges learned.
- C. Natural law only (thrust known)** T , the throttle profile and the edges are given: $\dot{a} - T \theta B = -c_d D$ over all samples (burns included) gives c_d .
- D. Both, learned separately** c_d from the clean *coast* arcs only, then T from the clean *burn* arcs given that c_d .

The forecast integrates the hysteresis law forward from the end of the last training burn with the learned rates (shapes averaged over the training coast and burn samples, i.e. with the measured, not circular, speeds) and compares the predicted burn onsets with the true ones over 360 days.

15.3 Results

scenario	c_d/true (BINDy $\pm 1\sigma$)	T/true	period [d]	onset error at 360 d
A whole system	0.9922 ($\pm 1.4\%$)	0.924	3.9351	1668 min
B thrust only (natural known)	given	0.924	3.9047	26 min
C natural only (thrust known)	1.0015 ($\pm 1.0\%$)	given	3.9006	302 min
D both, separately	1.0006 ($\pm 0.6\%$)	0.9995	3.9011	262 min
truth	1	1	3.9051	—

- **The joint regression is biased.** The GMM’s burn flags are slightly wider than the thrust (they include the ramps, and the smoothed derivative spreads the burn edge over the filter window), so in scenario A some samples with partial thrust are labelled “burn” and some with thrust are labelled “coast”. Least squares then lowers T by 7.6% and absorbs part of the edge leakage into the drag (-0.8%). Splitting the problem onto clean arcs (D) removes both biases: c_d to 0.06%, T to 0.05%.
- **The forecast is governed by the natural law.** A coast is 99.6% of each 3.9-day cycle, so a 0.1% drag error shifts every burn by about 5 minutes and accumulates linearly. With the natural law known (B) the forecast stays within 26 minutes over a year, even though its T is 7.6% low (that only lengthens each 23-minute burn by about 2 minutes). With the thrust known (C) the drag is learned to 0.15% and the forecast is far better than A but worse than B.
- **Rate errors.** Predicted $\dot{a}$ on coasting samples is within 0.001–0.04 m/hour of the truth for all scenarios; over all samples the error (about 20 m/hour) is dominated by the few samples at the burn edges where the detected flags and the true throttle disagree — except in C, which uses the true throttle (0.008 m/hour).

What prior knowledge is worth

For forecasting a station-keeping satellite, knowing the *natural* dynamics (a good drag model) is worth far more than knowing its thruster: the thrust level hardly matters, the coast rate decides when the next burn comes. When nothing is known, learn the two laws *separately* on clean arcs rather than in one regression.

Step 19: thrust law vs natural dynamics, learned together or apart, and with no deadband

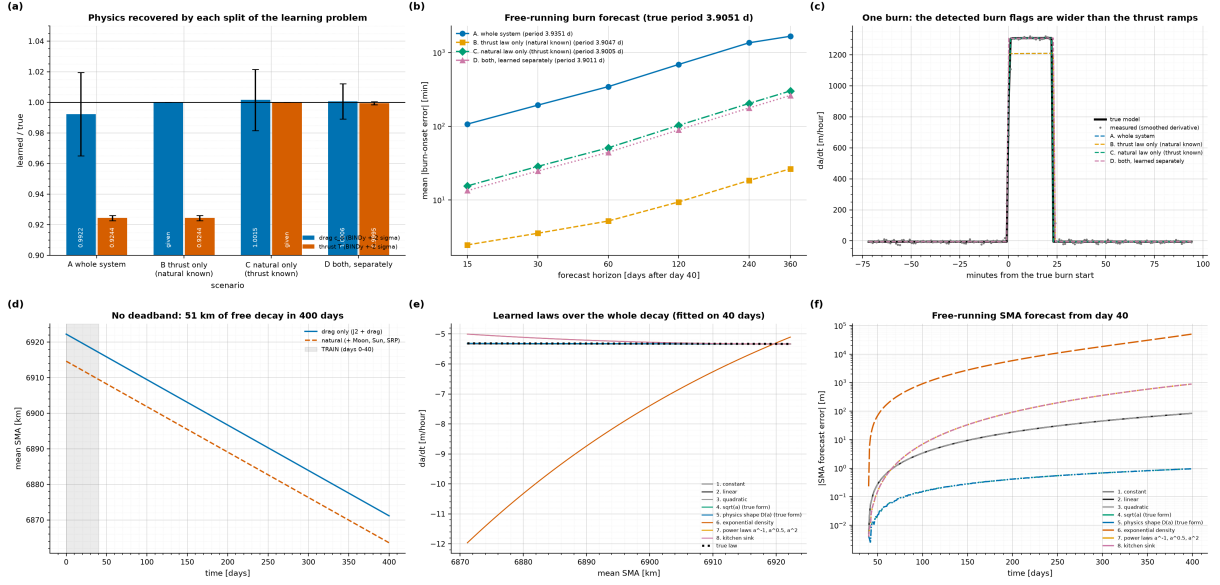


Figure 13: Step 19: (a) physics recovered per scenario with BINDy intervals, (b) free-running forecast error, (c) one burn as seen by each scenario, (d) the free decay without a deadband, (e) the learned decay laws, (f) their SMA forecasts.

15.4 The natural law with no deadband

Without the controller the satellite decays freely: 51 km in 400 days for both the drag-only run ($J_2 + \text{drag}$) and the natural run (plus Moon, Sun and SRP). With constant density the true law is

$$\dot{a} = -c_d \frac{2a^2}{\mu} |\mathbf{v}|^3 = -2c_d \sqrt{\mu a} \quad (\text{circular orbit}). \quad (23)$$

Eight libraries were fitted on days 0–40 (one SMA sample per minute, reference noise, 1-hour smoothing derivative) and tested on days 40–400:

library	test RMS of $\dot{a}$ [m/h]	SMA error after 360 d	recovered c_d
constant / linear / quadratic*	0.012	84 m	—
$\sqrt{a}$ (true form)	0.0006	1 m	1.0001
physics shape $D(a)$ (true form)	0.0006	1 m	1.0001
exponential density $e^{-(a-a_0)/H}$	3.55	51 km	—
power laws $a^{-1}, a^{1/2}, a^2$	0.14	0.9 km	—
kitchen sink	0.14	0.9 km	—

* STLSQ prunes the linear and quadratic terms, leaving a constant rate.

- The **true form** recovers c_d to 0.01% from 40 days and forecasts the SMA a year ahead to 1 m (also with the Moon, Sun and SRP present: their effect on the mean SMA is periodic).
- A **constant** rate is almost as good over 40 days but misses the 0.4% change of the rate over

the 51 km descent: 84 m after a year. STLSQ correctly prunes the linear and quadratic terms — over 40 days they cannot be distinguished from noise.

- The **exponential-density** library encodes physics that is *not* in this simulator (density rising as the orbit falls; here density is constant): it fits the training data but its extrapolation is 51 km off — a physically plausible library is not automatically the right one, and free-running forecasts, not training fit, decide.
- Collinear **power-law** and kitchen-sink libraries split the rate between near-identical columns and extrapolate to 0.9 km error.

In the real atmosphere the density *does* rise as the orbit decays (and varies with space weather, step 10), so on real data the exponential library would be expected to win; the method, not the winner, is the transferable result.

16 Validation, figure audit and honest limitations

16.1 Testing

41 unit tests (`python -m unittest discover tests`) cover the physics ($J_2 = -\nabla U_{J_2}$), the controller, finite differences, the bias–variance identity, the mean-element calibration, forecasting with perfect knowledge, conformal coverage, P_c against Monte Carlo, pass orbit determination, the OEM reader, regimes, fleet change detection, GEO law recovery and the SINDy zoo.

16.2 Problems found and fixed during the audits

- the J_2 reference radius (0.22% too weak);
- a J_2 check that ignored the drag-driven change of a (1.4% apparent error);
- a GMM burn component not centred on the burns (2nd-order derivative truncation error);
- a single tuned stencil order applied to 10-min data (10% c_d error, now per step);
- an over-smoothed stacked burn in step 8 (now its own tuned window);
- burn edges biased by partially-thrusting samples (now from coast/burn line intersections);
- a GP detector that used true labels (now GMM labels);
- integrators overshooting the band edges (now stopped on the edge).

16.3 Limitations

- The inputs of steps 1–9 are full state vectors at 6 s with 0.1 m / 0.5 mm/s noise — far better than real tracking (step 11 shows how quickly detection degrades with realistic passes).
- Constant drag in steps 1–9 makes the periodic baseline look unbeatable; step 10 corrects this with modelled, not real, space weather.
- Steps 10–16 use the mean-element model (burn period within 0.01% of the full run), not full orbit integration.
- No real ephemerides were downloaded (a space-track account is needed); optical angles-only orbit determination is not implemented.
- The per-step stencil order was tuned for c_d under noise; at the 2-min step it picked order 2, which is worse for μ on clean data (step 2a) — tuning is objective-specific.
- Moon, Sun and SRP strengths are only weakly identifiable from short or coarse records.

17 How well the project answers the brief

Brief requirement	Status
Figure 1	reproduced (step 5)
Detect past manoeuvres	92/92 burns, 0 false events over 360 days at reference noise; down to 10 m-noise tracking passes
Reconstruct the control law	edges to ~ 0.3 m, drag and thrust to $< 0.05\%$, thrust profile, memory (hysteresis), multiple regimes, GEO law
Predict future manoeuvres	yes; beats a periodic baseline once drag varies, with calibrated intervals
Screening / association / tasking	demonstrated quantitatively (steps 12–13) in a simplified setting
Real data, realistic noise	partly: tracking passes and an OEM reader; no real ephemerides yet

A Running the code

```
python main.py                # all 31 stages -> outputs/ (~30 min)
python main.py --list         # the stages, in run order
python main.py --stages step6 gmm # only some stages
python main.py --retune        # redo the step 0 bias-variance studies
python -m unittest discover tests # 41 tests
```

Library code is in `deadband/` (no plotting); each stage in `stages/` draws figures into `outputs/report/stepN_*/` and returns numbers for `outputs/results.json`. Every module starts with a plain-English docstring explaining its method.

B Glossary

SMA

semi-major axis; *osculating* from the two-body energy, *mean* including the J_2 potential.

Deadband

the allowed SMA band; the controller burns only at its edges.

Hysteresis

dependence of the state on its history (the controller’s memory).

GMM

Gaussian mixture model; EM = expectation–maximisation.

Event F1

harmonic mean of event precision and recall.

SINDy / STLSQ

sparse identification of nonlinear dynamics / sequentially thresholded least squares.

BINDy / ARD

the Bayesian variant / automatic relevance determination.

EKF

extended Kalman filter; STM = state-transition matrix.

TCA, P_c

time of closest approach, collision probability.

Conformal prediction

distribution-free intervals calibrated on held-out errors.

OEM

CCSDS orbit ephemeris message (ephemeris file format).